GENERAL CHEMISTRY

Chapter 7

Properties of Solutions

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The Solution

- ✓ Dissolution of solid in liquids
- ✓ Dissolution of liquids in liquids
- ✓ Dissolution of gases in liquids
- ✓ Effect of temperature on solubility
- ✓ Effect of pressure on solubility
- ✓ Concentration
- ✓ Boiling point elevation
- ✓ Freezing Point depression
- ✓ Osmotic Pressure

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The Solution

- A solution is a homogeneous mixture of solute (present in smallest amount) and solvent (present in largest amount).
  - ✓ Solution may be gas, liquid, or solids.
  - ✓ Solvent is the substances used to dissolve the solute.

TABLE 13.1 Examples of Solutions

State of Solution	State of Solvent	State of Solute	Example
Gas	Gas	Gas	Air
Liquid	Liquid	Gas	Oxygen in water
Liquid	Liquid	Liquid	Alcohol in water
Liquid	Liquid	Solid	Salt in water
Solid	Solid	Gas	Hydrogen in palladium
Solid	Solid	Liquid	Mercury in silver
Solid	Solid	Solid	Silver in gold

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AQUEOUS SOLUTIONS

- The solvent is generally in excess.
- Example:

The solution NaCl(aq) is sodium chloride NaCl(s) dissolved in water $H_2O(l)$.

The solute is NaCl(s) and the solvent is $H_2O(l)$.

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Saturated solution

- **Saturated** means that the maximum amount is dissolved in the solution, under normal conditions.
- **Dissolve:** solute + solvent → solution
- In a **saturated solution**, the solvent contains the maximum amount of solute.
- **Example:** At 90°C, 40 g of NaCl(s) in 100g H_2O(l) represent a saturated solution.

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Unsaturated solution

- In an **unsaturated solution**, the solvent contains less than the maximum amount of solute.
- Example:
  - ✓ At 90°C, 30 g of NaCl(s) in 100g H_2O(l) represent an unsaturated solution.
  - ✓ If adding 10 g of NaCl(s) into that solution, the saturated solution will be form.

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Supersaturated solution

- In a **supersaturated solution**, the solvent contains more than the maximum amount of solute.
- A supersaturated solution is very unstable and the amount in excess can precipitate or crystallize.
- **Crystallization**: solution → solute + solvent
- Example:

At 90°C, 50 g of NaCl(s) in 100g H_2O(l) represent a supersaturated solution. Eventually, 10 g of NaCl(s) will precipitate.

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Solubility

- **Solubility**: amount of solute required to form a saturated solution.
- The solubility of a solute in a given amount of solvent is dependent on:
  - ✓ The temperature,
  - ✓ The pressure,
  - ✓ And the chemical natures of the solute and solvent.

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Effect of Temperature on Solubility

- ✓ In general, as the temperature of a solution increases the solubility increases.
- ✓ Example: Increasing the solution temperature allows more sugar to go into solution.
- ✓ Therefore, it is an endothermic process (heat is on the reactant side).

$$\text{heat} + \text{sugar (s)} \rightleftharpoons[H_2O] \text{sugar (aq)}$$

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Effect of Temperature on Solubility

❖ **Solid in liquid:**

- ✓ **Endothermic dissolving:** Solubility increases with increasing temperature.
- ✓ **Exothermic dissolving:** solubility decreases with increasing temperature.

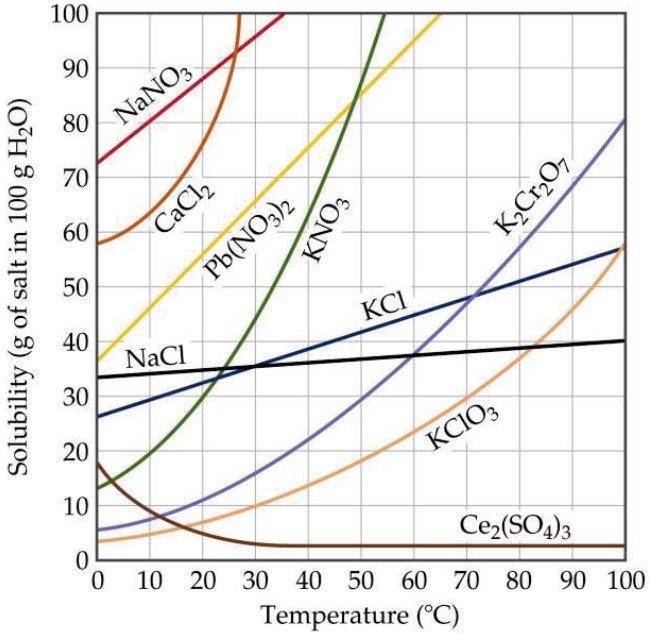
❖ Many ionic solids dissolve by endothermic processes.
→ As temperature increases, solubility of solids generally increases.

❖ For example, KCl dissolves endothermically.

❖ Sometimes, solubility decreases as temperature increases (e.g. $Ce_2(SO_4)_3$, Cerous Sulfate).

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Effect of Temperature on Solubility



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Effect of Temperature on Solubility

❖ **Liquid in liquid:**

- ✓ For partially dissolving liquids like dimethyl ether(CH_3-O-CH_3) in water (H_2O), solubilty increases with increasing temperature.
- ✓ For a completely dissolving liquids like ethyl alcohol (C_2H_5OH) in water(H_2O), solubility decreases with increasing temperature.

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Effect of Temperature on Solubility

- **Gas in liquid:**
  - ✓ Solubility of a gas in a liquid almost always decreases with increasing temperature.
  - ✓ ***The reason:*** Increased temperature causes an increase in kinetic energy. The higher kinetic energy causes more motion in molecules which break intermolecular bonds and escape from solution.

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Effect of Gas Pressure on Solubility

- **Liquids** and **solids** exhibit practically no change of solubility with changes in pressure.
- The solubility of a **gas** increases as the pressure increases.
  - ✓ *If the pressure is increased, the gas molecules are "forced" into the solution.*
  - ✓ *The number of gas molecules dissolved in solution has increased.*

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Effect of Gas Pressure on Solubility

- All carbonated beverages are bottled under pressure to increase the carbon dioxide dissolved in solution.
- When the bottle is opened, the pressure above the solution decreases. As a result, the solution effervesces and some of the carbon dioxide bubbles off.
- **Quiz:** Champagne continues to ferment in the bottle. The fermentation produces CO_2. Why is the cork wired on a bottle of champagne?
- **Answer:** As more CO_2 is formed , the pressure of the gas increase.The wire is to prevent the cork from blowing off.

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The rate of solution

- The rate of solution is a measure of how fast a substance dissolves. Some of the factors determining the rate of solution are:

✓ **Size of the particles:** When a solute dissolves, the action takes place only at the surface of each particle. When the total surface area of the solute particles is increased, the solute dissolves more rapidly.

✓ → Breaking a solute into smaller pieces increases its surface area and hence its rate of solution.

✓ **Stirring:** With liquid and solid solutes, stirring brings fresh portions of the solvent in contact with the solute, thereby increasing the rate of solution.

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The rate of dissolving

- Amount of solute already dissolved:
  - ✓ When there is little solute already in solution, dissolving takes place relatively rapidly.
  - ✓ As the solution approaches the point where no solute can be dissolved, dissolving takes place more slowly.
- Temperature: For solid, liquid and gaseous solutes, changing the temperature not only changes the amount of solute that will dissolve but also changes the rate at which the solute will dissolve.

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Chemical natures of the solute and solvent

- A polar solute will dissolve in a polar solvent but not in a nonpolar solvent. The adage "like dissolves like" is very useful.
  - ✓**Polar solute/polar solvent:** ethanol, salt, sugar in water**.**
  - ✓**Nonpolar solute/nonpolar solvent:** Iodine in carbontetrachloride, gasoline or benzene.
- Example:
  - ✓Alcohol dissolves in water.
  - ✓Water does not dissolve in oil.

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Concentration

- Mass Percentage
- ppm
- ppb
- Mole Fraction
- Molarity
- and Molality

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Ways of Expressing Concentration

$$\text{mass \% of component} = \frac{\text{mass of component in solution}}{\text{total mass of solution}} \times 100$$

$$\text{ppm of component} = \frac{\text{mass of component in solution}}{\text{total mass of solution}} \times 10^6$$

- Parts per million (ppm) can be expressed as 1 mg of solute per kilogram of solution.
  - If the density of the solution is 1g/mL, then 1 ppm = 1 mg solute per liter of solution.

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Concentration

$$\text{ppb of component} = \frac{\text{mass of component in solution}}{\text{total mass of solution}} \times 10^9$$

- Parts per billion (ppb) are 1 μg of solute per kilogram of solution.

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Concentration

$$\text{Mole fraction of component} = \frac{\text{moles of component in solution}}{\text{total moles of solution}}$$

$$\text{Molarity} = \frac{\text{moles solute}}{\text{liters of solution}}$$

$$\text{Molality}, m = \frac{\text{moles solute}}{\text{kg of solvent}}$$

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Practice Problems

EXAMPLE 14-3 ***Mole Fraction***

What are the mole fractions of CH_3OH and H_2O in the solution described in Example 14-1? It contains 128 grams of CH_3OH and 108 grams of H_2O.

Solution

$$X_{CH_3OH} = \frac{\text{no. mol } CH_3OH}{\text{no. mol } CH_3OH + \text{no. mol } H_2O} = \frac{4.00 \text{ mol}}{(4.00 + 6.00) \text{ mol}} = 0.400$$

$$X_{H_2O} = \frac{\text{no. mol } H_2O}{\text{no. mol } CH_3OH + \text{no. mol } H_2O} = \frac{6.00 \text{ mol}}{(4.00 + 6.00) \text{ mol}} = 0.600$$

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Practice Problems

EXAMPLE 14-1 ***Molality***

What is the molality of a solution that contains 128 g of CH_3OH in 108 g of water?

Solution

$$\frac{? \text{ mol } CH_3OH}{\text{kg } H_2O} = \frac{128 \text{ g } CH_3OH}{0.108 \text{ kg } H_2O} \times \frac{1 \text{ mol } CH_3OH}{32.0 \text{ g } CH_3OH} = \frac{37.0 \text{ mol } CH_3OH}{\text{kg } H_2O}$$

$$= 37.0\ m\ CH_3OH$$

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Practice Problems

EXAMPLE 14-2 ***Molality***

How many grams of H_2O must be used to dissolve 50.0 grams of sucrose to prepare a 1.25 *m* solution of sucrose, $C_{12}H_{22}O_{11}$?

Solution

$$\text{kg } H_2O = \frac{\text{mol } C_{12}H_{22}O_{11}}{\text{molality of solution}} = \frac{0.146 \text{ mol } C_{12}H_{22}O_{11}}{1.25 \text{ mol } C_{12}H_{22}O_{11}/\text{kg } H_2O}$$

$$= 0.117 \text{ kg } H_2O = 117 \text{ g } H_2O$$

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Practice Problems

1) A solution containing equal masses of glycerol, $C_3H_8O_3$, and water has a density of 1.10g/mL. Calculate the: (a) Molarity, (b) Mole Fraction of glycerol, (c) Molality of the solution.

2) Calculate the percent by mass of $CaCl_2$ in a solution containing 5.2g $CaCl_2$ in 450g of water.

3) Calculate the ppm, by mass, of $CaCl_2$ in a solution containing 0.149 moles of $CaCl_2$ in 443g of water.

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Practice Problems

1) Calculate the molarity when 0.020 moles of glycerol, $C_3H_8O_3$, is dissolved in 50 g of water at room temperature. Water has a density of 1.00g/mL.

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Practice Problem

1)A 0.500 L hydrochloric acid (HCl) solution, which has a density of 1.10 g/mL, contains 36% HCl by mass. Calculate the mole fraction of HCl.

2) A solution containing equal masses of NaCl and water has a density of 1.10g/mL. Calculate the molality of the solution.

3) Calculate the ppm, by mass, of $CaCl_2$ in a solution containing 0.149 moles of $CaCl_2$ in 443g of water.

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Solubility of a Gas

Pressure Effects:

- Henry's Law:

$$S_g = kP_g$$

- Where:

  S_g is the solubility of a gas,

  k is a constant,

  and P_g is the partial pressure of a gas.

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Raoult's Law

❖ **Raoult's Law:**

$$P_{solvent} = X_{solvent} P^{O}_{solvent}$$

where:

$X_{solvent}$ *: the mole fraction of the solvent in a solution.*

P^o *is the vapor pressure of the pure solvent,*

$P_{solvent}$ *is the vapor pressure of the solvent in the solution.*

❖ **The lowering of the vapor pressure, $\Delta P_{solvent}$,**

$$\Delta P_{solvent} = P^0_{solvent} - P_{solvent}$$

$$\Delta P_{solvent} = X_{solute} P^0_{solvent}$$

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Raoult's Law

EXAMPLE 14-4 *Vapor Pressure of a Solution of Nonvolatile Solute*

Sucrose is a nonvolatile, nonionizing solute in water. Determine the vapor pressure lowering, at 25°C, of the 1.25 *m* sucrose solution in Example 14-2. Assume that the solution behaves ideally. The vapor pressure of pure water at 25°C is 23.8 torr (Appendix E).

Solution

$$X_{\text{sucrose}} = \frac{0.146 \text{ mol}}{0.146 \text{ mol} + 6.50 \text{ mol}} = 0.0220$$

$$\Delta P_{\text{solvent}} = (X_{\text{solute}})(P^0_{\text{solvent}}) = (0.0220)(23.8 \text{ torr}) = 0.524 \text{ torr}$$

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Raoult's Law

EXAMPLE 14-5 *Vapor Pressure of a Solution of Volatile Components*

At 40°C, the vapor pressure of pure heptane is 92.0 torr and the vapor pressure of pure octane is 31.0 torr. Consider a solution that contains 1.00 mole of heptane and 4.00 moles of octane. Calculate the vapor pressure of each component and the total vapor pressure above the solution.

Solution

$$X_{\text{heptane}} = \frac{1.00 \text{ mol heptane}}{(1.00 \text{ mol heptane}) + (4.00 \text{ mol octane})} = 0.200$$

$$X_{\text{octane}} = 1 - X_{\text{heptane}} = 0.800$$

$$P_{\text{heptane}} = X_{\text{heptane}} P^0_{\text{heptane}} = (0.200)(92.0 \text{ torr}) = 18.4 \text{ torr}$$

$$P_{\text{octane}} = X_{\text{octane}} P^0_{\text{octane}} = (0.800)(31.0 \text{ torr}) = 24.8 \text{ torr}$$

$$P_{\text{total}} = P_{\text{heptane}} + P_{\text{octane}} = 18.4 \text{ torr} + 24.8 \text{ torr} = 43.2 \text{ torr}$$

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Boiling-Point Elevation

- At the normal boiling point of pure liquid, the vapor pressure the solution will be < 1atm. Therefore, a higher temperature, ΔT_b, is required to attain a vapor pressure of 1 atm for the solution.
- **Molal boiling-point-elevation constant**, K_b, expresses how much ΔT_b changes with molality, m:

$$\Delta T_b = K_b m$$

- K_b is dependent only on the solvent used in the making of the solution.

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Boiling-Point Elevation

TABLE 14-2 *Some Properties of Common Solvents*

Solvent	bp (pure)	K_b (°C/m)	fp (pure)	K_f (°C/m)
water	100*	0.512	0*	1.86
benzene	80.1	2.53	5.48	5.12
acetic acid	118.1	3.07	16.6	3.90
nitrobenzene	210.88	5.24	5.7	7.00
phenol	182	3.56	43	7.40
camphor	207.42	5.61	178.40	40.0

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Freezing Point Depression

- The freezing point is the temperature at which the first crystal forms.
- The change in freezing point can be defined by:

$$\Delta T_f = K_f \times m$$

- K_f is the molal freezing-point-depression constant.
- K_f is dependent only on the solvent used in the making of the solution.

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Freezing Point Depression

EXAMPLE 14-9 ***Freezing Point Depression***

Calculate the freezing point of the 1.25 *m* sucrose solution in Example 14-2.

Solution

From Table 14-2, K_f for H_2O = 1.86°C/*m*, so

$$\Delta T_f = (1.86°C/m)(1.25\ m) = 2.32°C$$

The solution freezes at a temperature 2.32°C *below* the freezing point of pure water,

$$T_{f(solution)} = 0.00°C - 2.32°C = -2.32°C$$

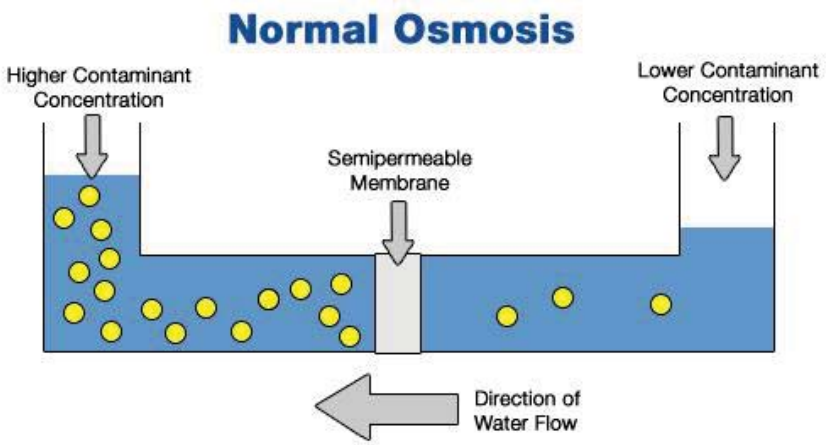
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Osmosis

- **Osmosis:** the movement of a solvent from low solute concentration to high solute concentration.
- **Semipermeable membrane**: permits passage of some components of a solution. Example: cell membranes
- There is movement in both directions across a semipermeable membrane.
- As solvent moves across the membrane, the fluid levels becomes uneven.

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Osmosis



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Osmotic Pressure

- Osmotic pressure, π, is the pressure required to stop osmosis:

$$\pi V = nRT$$

$$\pi = \left(\frac{n}{V}\right)RT$$

$$= MRT$$

- Isotonic solutions: two solutions with the same π separated by a semipermeable membrane.
- Hypotonic solutions: a solution of lower π with respects to the more concentrated solution.
- The osmotic process is spontaneous.

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Osmotic Pressure

EXAMPLE 14-14 ***Molecular Weight from Osmotic Pressure***

Pepsin is an enzyme present in the human digestive tract. A solution of a 0.500-gram sample of purified pepsin in 30.0 mL of aqueous solution exhibits an osmotic pressure of 8.92 torr at 27.0°C. Estimate the molecular weight of pepsin.

Solution

$$\pi = MRT = \left(\frac{n}{V}\right)RT \qquad \text{or} \qquad n = \frac{\pi V}{RT}$$

We convert 8.92 torr to atmospheres to be consistent with the units of R.

$$n = \frac{\pi V}{RT} = \frac{\left(8.92 \text{ torr} \times \dfrac{1 \text{ atm}}{760 \text{ torr}}\right)(0.0300 \text{ L})}{\left(0.0821 \dfrac{\text{L·atm}}{\text{mol·K}}\right)(300 \text{ K})} = 1.43 \times 10^{-5} \text{ mol pepsin}$$

Thus, 0.500 g of pepsin is 1.43×10^{-5} mol. We now estimate its molecular weight.

$$? \text{ g/mol} = \frac{0.500 \text{ g}}{1.43 \times 10^{-5} \text{ mol}} = 3.50 \times 10^{4} \text{ g/mol}$$

The molecular weight of pepsin is approximately 35,000 amu. This is typical for medium-sized proteins.

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THE pH AND pOH SCALES

- The pH and pOH scales provide a convenient way to express the acidity and basicity of dilute aqueous solutions. The pH and pOH of a solution are defined as

$$pH = -\log [H_3O^+] \quad \text{or} \quad [H_3O^+] = 10^{-pH}$$
$$pOH = -\log [OH^-] \quad \text{or} \quad [OH^-] = 10^{-pOH}$$

$$[H_3O^+][OH^-] = 1.0 \times 10^{-14}$$

and $pH + pOH = 14.00$ (at 25°C)

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THE pH AND pOH SCALES

EXAMPLE 18-5 ***Calculations Involving pH and pOH***

Calculate $[H_3O^+]$, pH, $[OH^-]$, and pOH for a 0.015 *M* HNO_3 solution.

EXAMPLE 18-6 ***Calculations Involving pH and pOH***

Calculate $[H_3O^+]$, pH, $[OH^-]$, and pOH for a 0.015 *M* $Ca(OH)_2$ solution.

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